

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Cummins® electronic service tool, or equivalent.

Additional Service Items

- Shop air
- Soapy water

General Information

This document contains information concerning non self-calibrating Variable Geometry Turbocharger (VGT) actuators and self-calibrating VGT actuators.

To identify the VGT actuator type, use Service Bulletin, Self-Calibrating Variable Geometry Turbocharger (VGT) Actuator, Bulletin 6467611 (</qs3/pubsys2/xml/en/bulletin/6467611.html>).

A video supports the information provided in this procedure.

Read the entire procedure prior to viewing the video.

Preparatory Steps

WARNING

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

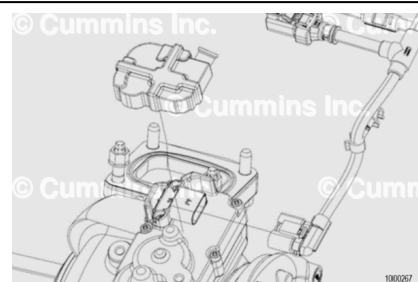
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Clean the area around the VGT actuator and dry with compressed air.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html>)
- Disconnect the VGT actuator coolant return lines from the VGT actuator. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-041.html>)

Remove

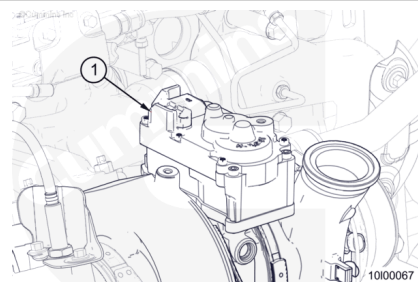
Disconnect the wiring harness from the VGT actuator by pushing down on the release lever and pulling the connection apart.

If equipped, the VGT actuator connector cover wire tie does **not** need to be removed to be disconnected. Carefully move the cover out of the way to access the connector release.



⚠ CAUTION ⚠

Verify the coolant has been drained before the VGT actuator is removed. Failure to do so can result in damage to the VGT actuator.



Remove VGT actuator mounting capscrews.

Remove the VGT actuator (1).

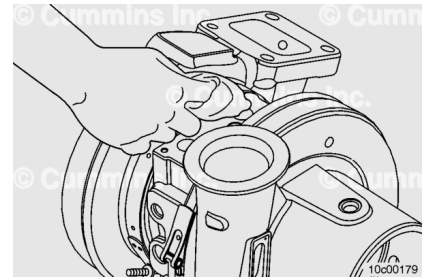
Remove o-rings and inspect for reuse.

Clean and Inspect for Reuse

Use a clean cloth to wipe around the surface where the VGT actuator seals to the VGT.

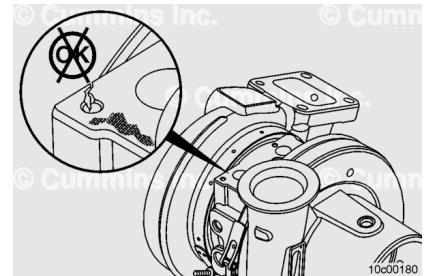
Take care **not** to drop any dirt or debris into the VGT.

Inspect the sealing surfaces for signs of leaks, cracks or other damage.



If the sealing surface of the turbocharger is damaged and causing a leak, the VGT **must** be replaced. Discontinue work with this procedure. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-033.html>)

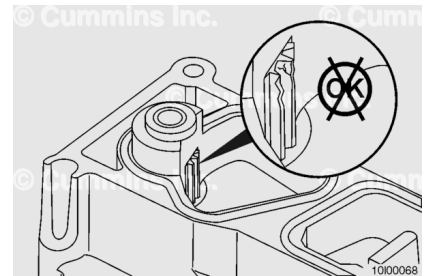
If the sealing surface of the VGT actuator is causing a leak, replace **only** the VGT actuator.



Inspect the pinion gear on the VGT actuator for excessive wear, or damaged teeth.

Replace the VGT actuator if the teeth are excessively worn or damaged.

Note : It is extremely difficult to move the pinion gear by hand; therefore, moving the pinion gear by hand is **not** recommended to check for the functionality.

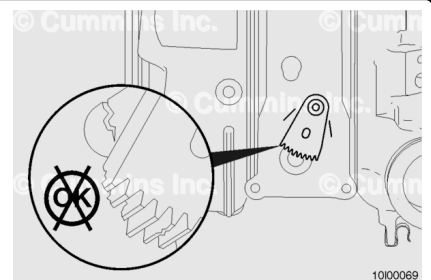


⚠ CAUTION ⚠

Do not use any hand tools to rotate the turbocharger sector gear or damage to the turbocharger sector gear could result.

Inspect the turbocharger sector gear on the VGT for excessive wear, damaged teeth, or a broken shaft.

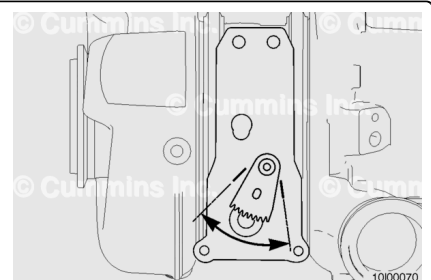
Replace the VGT if the turbocharger sector gear shows signs of damage.



Grasp the turbocharger sector gear by hand and move the turbocharger sector gear through its operational range.

Within the VGT there are solid end stops which limit turbocharger sector gear travel in both directions.

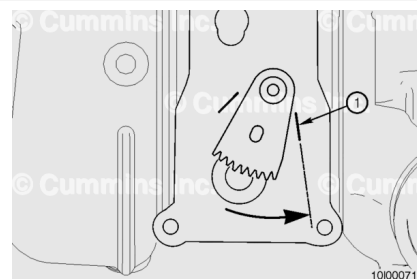
Note : The turbocharger sector gear may take considerable force to move. Moving the turbocharger sector gear by hand is considered acceptable.



The turbocharger sector gear being difficult to move is **not** a reason to replace the turbocharger. Do **not** replace the turbocharger if the turbocharger sector gear makes full travel from stop to stop.

Rotate the turbocharger sector gear **counterclockwise**, toward the VGT compressor housing, until the turbocharger sector gear stops.

The edge of the turbocharger sector gear **must** touch the engraved bearing housing hash mark (1).

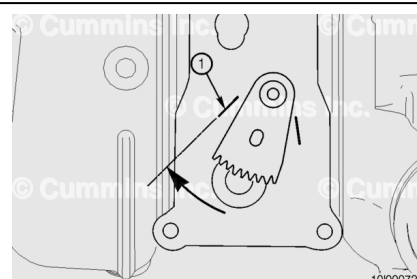


Rotate the turbocharger sector gear **clockwise**, toward the VGT turbine housing, until the turbocharger sector gear stops. The edge of the turbocharger sector gear **must** touch the engraved bearing housing hash mark (1).

If the turbocharger sector gear does **not** go through the full range of motion, replace the VGT.

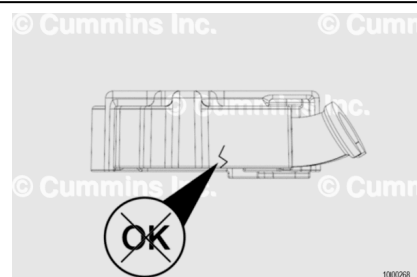
Refer to Procedure 010-033 in Section 10.

(/qs3/pubsys2/xml/en/procedures/377/377-010-033.html)



Inspect the VGT Actuator Harness Cover.

- Inspect the cover for damage.
- Do **not** remove the wire tie for inspection.
- Replace the cover if damage is found.



Install

⚠ CAUTION ⚠

During the VGT actuator installation procedure, the pinion gear on the VGT actuator may move. Keep hands and tools away from the pinion gear during the procedure.



A video supports the information provided in this procedure. Read the procedure prior to viewing the video.

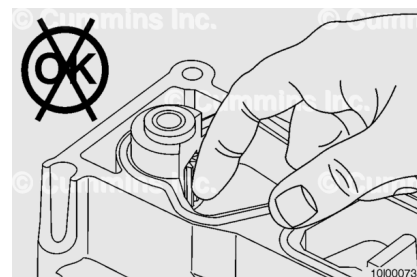
Without Self-Calibrating Turbocharger Actuator

New VGT actuators are sold with a calibration software installed. Downloading a new calibration software into the VGT actuator should only be required if Fault Code 4956 or 4957 is active after installing a new VGT actuator, or if directed by a Campaign, Temporary Repair Practice, or warrantable repair.

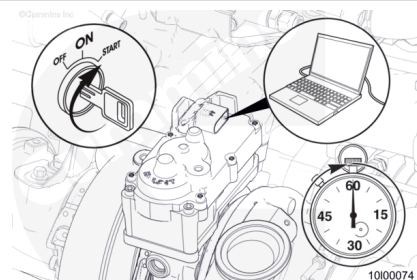
If the VGT actuator requires a calibration code to be downloaded, use the following procedure. Refer to Procedure 010-150 in Section 10.

(/qs3/pubsys2/xml/en/procedures/377/377-010-150.html)

Following these instructions in order is very important.



- Continue through the entire VGT actuator installation procedure before attempting to troubleshoot any other fault codes.
- Verify that the VGT actuator is removed from the turbocharger bearing housing.
- Verify that the VGT actuator electrical connector is disconnected from the engine wiring harness.
- Turn the keyswitch ON. Connect INSITE™ electronic service tool and wait 60 seconds.



The keyswitch **must** remain in the ON position throughout both the Install and Calibrate procedures unless otherwise directed.

- Connect the VGT actuator electrical connector to the engine wiring harness.

If Fault Code 1897 becomes active, do the following.

- Disconnect the VGT actuator connector from the engine wiring harness with the keyswitch ON.
- Connect the VGT actuator electrical connector. Fault Code 1897 will go inactive.
- Active Fault Code(s) 2449 or 6432 are normal and expected when a new VGT actuator is connected to the engine, because the VGT actuator is **not** calibrated to the turbocharger.

Continue through the engine VGT actuator installation procedure before attempting to troubleshoot any other fault codes.

Use INSITE™ electronic service tool and go to the engine control module (ECM) Diagnostic Tests screen.

From the list, select VGT Electronic Actuator Installation and Calibration and click on the “Next” button.

Note : The VGT Electronic Actuator Installation and Calibration is not a diagnostic test. This procedure is used to properly install and calibrate the VGT actuator. Running this procedure improperly can result in additional fault codes and/or damage to the turbocharger or VGT actuator.

INSITE™ electronic service tool “INSTALL ACTUATOR” command **must** be run with the VGT actuator **not** mounted to the turbocharger. The "Install Actuator" procedure will **not** be successful if run with the VGT actuator mounted on the turbocharger.

Locate the column labeled “Value” and left click on the down arrow. Select “INSTALL ACTUATOR” then select “START”.

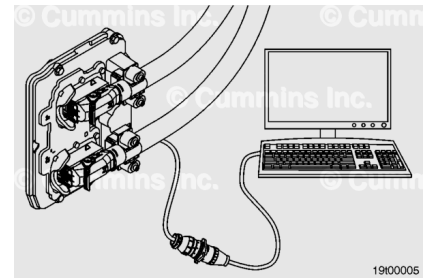
The VGT actuator pinion gear will be set to a known position to prepare the VGT actuator pinion gear for installation to the turbocharger. This step should take less than 30 seconds to complete with INSITE™ electronic service tool.

INSITE™ electronic service tool will indicate when this step is complete.

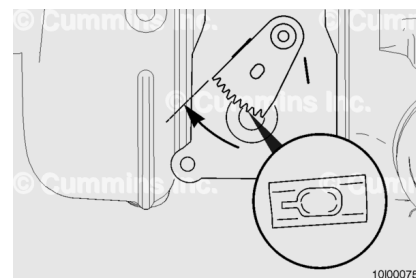
Fault Code(s) 2449 or 6432 will be active at this point in the procedure. Continue through the VGT actuator procedure before troubleshooting any fault codes.

If, at any point, an INSITE™ electronic service tool status message indicates the procedure was stopped or failed, leave the keyswitch ON and cycle the power to the VGT actuator by unplugging the VGT actuator from the harness, and then reconnecting it.

If cycling the power to the VGT actuator does **not** work, unplug the VGT actuator, turn the keyswitch OFF for 30 seconds, and then restart INSITE™ electronic service tool. Then start over, beginning with the “INSTALL ACTUATOR” step.



Grasp the turbocharger sector gear by hand and rotate the turbocharger sector gear **clockwise** toward the turbocharger turbine housing. Verify the edge of the turbocharger sector gear is rotated all the way toward engraved hash mark on the bearing housing.

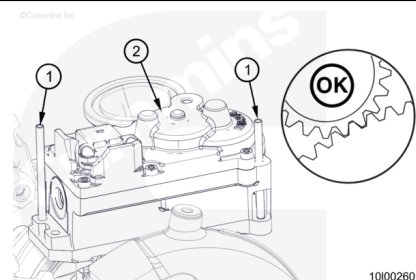


Use of the full amount of the Holset® supplied grease is critical for reliable operation of the turbocharger VGT actuator.

Coat the teeth on the turbocharger sector gear with the grease packet supplied in the installation kit.

⚠ CAUTION ⚠

Do not attempt to force the VGT actuator onto the bearing housing by using the capscrews. Misalignment can cause damage to the VGT actuator or turbocharger.



Verify o-rings are installed on the VGT actuator.

Insert the threaded guide pins into the diagonally opposite fastener holes in the bearing housing and locate the VGT actuator onto the guide pins.

Slide the VGT actuator into position, verify the pilot bearing is aligned with the mating hole in the bearing housing and the output gear of the VGT actuator is smoothly engaged with the turbocharger sector gear.

Insert the long and short VGT actuator socket head capscrews in the remaining fastener holes and hand-tighten.

The VGT actuator pinion gear and the turbocharger sector gear **must** engage smoothly.

With the VGT actuator aligned on the bearing housing, replace the guide pins with the VGT actuator capscrews.

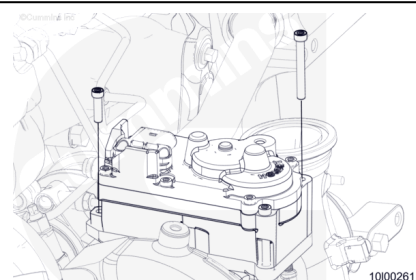
Install the remaining two capscrews.

Tighten all capscrews in a criss-cross pattern in two steps.

Torque Value:

1. 3 n•m [27 in-lb]

2. 11 n•m [97 in-lb]



With Self-Calibrating Turbocharger Actuator

Note : Additional information for self-calibrating VGT actuator is available for this unit. Reference Service Bulletin Self-Calibrating Variable Geometry Turbocharger (VGT) Bulletin 6467611 (/qs3/pubsys2/xml/en/bulletin/6467611.html).

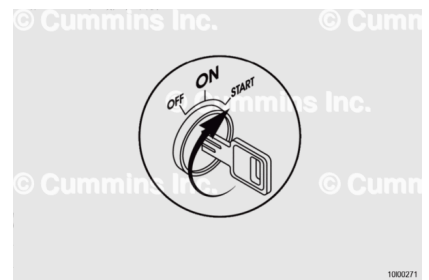
The VGT actuator **must** be below 50°C [122°F] for the self-calibration process to initiate.

Verify that the VGT actuator is removed from the VGT bearing housing.

Verify that the keyswitch is OFF and that no power is present at the VGT actuator connector.

Connect the VGT actuator electrical connector to the engine wiring harness. Place the VGT actuator in a secure location and verify nothing is touching the output gear.

Turn the keyswitch ON. Wait for 10 seconds. The VGT actuator gear will move until the VGT actuator has completed its functionality check.



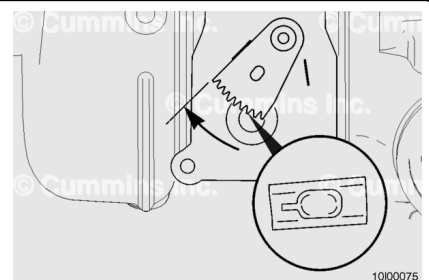
Turn the keyswitch OFF.

Active fault codes are normal and expected when a new VGT actuator is connected to the engine because the VGT actuator is **not** calibrated to the VGT. Continue through the entire VGT actuator installation procedure before attempting to troubleshoot any fault codes.

Grasp the turbocharger sector gear by hand and rotate the turbocharger sector gear **clockwise** toward the VGT turbine housing. Verify the edge of the turbocharger sector gear is rotated all the way toward engraved hash mark on the bearing housing.

Use of the full amount of the Holset® supplied grease is critical for reliable operation of the VGT actuator.

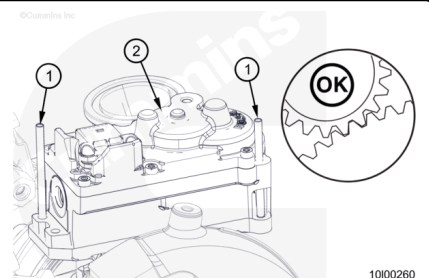
Coat the teeth on the turbocharger sector gear with the grease packet supplied in the installation kit.



⚠ CAUTION ⚠

Do not attempt to force the VGT actuator onto the bearing housing by using the capscrews. Misalignment can cause damage to the VGT actuator or turbocharger.

Verify o-rings are installed on the VGT actuator.



Insert the threaded guide pins into the diagonally opposite fastener holes in the bearing housing and locate the VGT actuator onto the guide pins.

Slide the VGT actuator into position, verify the pilot bearing is aligned with the mating hole in the bearing housing and the output gear of the VGT actuator is smoothly engaged with the VGT sector gear.

Insert the long and short VGT actuator socket head capscrews in the remaining fastener holes and hand-tighten.

The VGT actuator pinion gear and the VGT sector gear **must** engage smoothly.

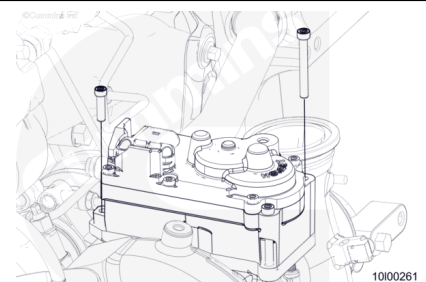
With the VGT actuator aligned on the bearing housing, replace the guide pins with the VGT actuator capscrews.

Install the remaining two capscrews.

Tighten all capscrews in a criss-cross pattern in two steps.

Torque Value:

1. 3 n•m [27 in-lb]
2. 11 n•m [97 in-lb]



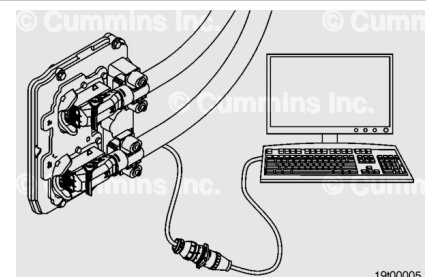
Calibrate

The VGT actuator **must** be calibrated to the VGT. This step **must** be performed to verify proper VGT operation is achieved.

Without Self-Calibrating Turbocharger Actuator

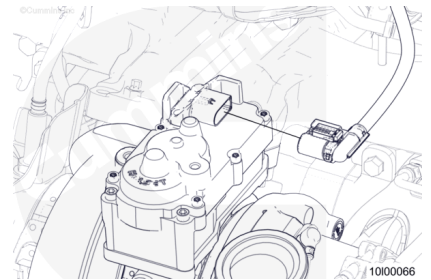
Calibration Instructions

- Within the INSITE™ electronic service tool screen labeled “VGT Electric Actuator Install and Calibrate”, locate the column labeled “Value” and left click on the down arrow. Select “CALIBRATE ACTUATOR”.
- INSITE™ electronic service tool “CALIBRATE ACTUATOR” command **must only** be run with the VGT actuator mounted to the turbocharger.
- Follow the instruction on the screen in order to calibrate the VGT actuator to the turbocharger. INSITE™ electronic service tool will indicate when this step is complete.



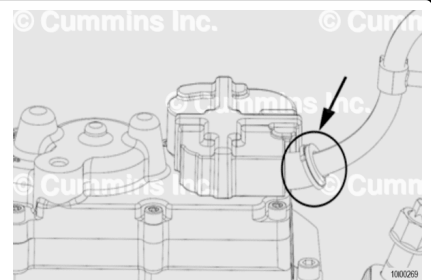
- If INSITE™ electronic service tool status message indicates the procedure was stopped or failed, turn the keyswitch OFF for 30 seconds and key back ON. Then start over, beginning with the “INSTALL ACTUATOR”.
- It is normal to have an active Fault Code 2387 at this point.
- Turn the keyswitch in the OFF position for 30 seconds. Then key ON and refresh the fault code screen. All VGT actuator fault codes should be inactive. Use INSITE™ electronic service tool to clear all fault codes.

Connect the VGT actuator connector.



Install the VGT actuator connector cover.

Install new wire tie if previously removed.



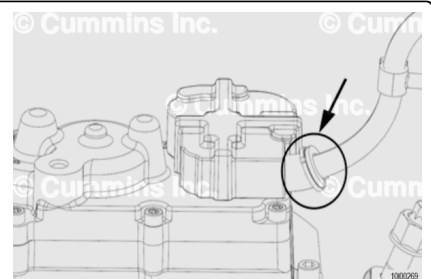
With Self-Calibrating Turbocharger Actuator

Calibration Instructions

- Turn the keyswitch ON. The VGT actuator will calibrate to the VGT. Calibration of VGT actuator takes about 10 seconds.
- After the VGT actuator has calibrated check for fault codes and follow the key on/off procedure to clear them. VGT actuator fault codes should go inactive after a successful install.

Install the VGT actuator connector cover.

Install new wire tie if previously removed.



Finishing Steps

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the VGT actuator coolant return lines. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-041.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html>)
- Operate the engine to verify the VGT actuator operates properly and all VGT actuator fault codes are inactive.
- Check the engine for leaks.

View video linked below for additional information on turbocharger actuator calibration and installation.

Note : http://tsb.cumminsvirtualcollege.com/en_647_010_134.aspx
(http://tsb.cumminsvirtualcollege.com/en_647_010_134.aspx)